QUESTION 1

(a) The two-dimensional x-component momentum equation reads as follows:

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)+\rho g_{\mathrm{x}}$$

[1 mark for each assumption – 5 marks in total]

- Steady-state: $\frac{\partial}{\partial t}=0$ [drop 1st term LHS]
- Flow is unidirectional: $u\neq 0,\ v=0$ [drop: 3rd term LHS]
- Flow is fully developed: $\frac{\partial u}{\partial x}=0$ [drop 2nd term LHS and 2nd term RHS]

The two-dimensional y-component momentum equation reads as follows:

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right)=-\frac{\partial p}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)+\rho g_{\mathrm{y}}$$

- Steady-state: $\frac{\partial}{\partial t}=0$ [drop 1st term LHS]
- Flow is unidirectional: $v=0,\ \frac{\partial v}{\partial x}=0,\ \frac{\partial v}{\partial y}=0$ [drop: 2nd and 3rd term LHS, 2nd and 3rd terms RHS]

The simplified equations read as follows: **[1 mark]**

For the x-component:

$$0=-\frac{\partial p}{\partial x}+\mu\frac{\partial^2 u}{\partial y^2}+\rho g_{\mathrm{x}}$$

For the y-component:

$$0=-\frac{\partial p}{\partial y}+\rho g_{\mathrm{y}}$$

(b) The following scalings are used: [**2 marks**]

$$\tilde{x}=\frac{x}{L}\quad \tilde{y}=\frac{y}{h}\quad \tilde{u}=\frac{u}{U}\quad \tilde{p}=\frac{p-p_0}{P}\quad \tilde{g}_x=\frac{g_x}{g}$$

The Reynolds and Froude numbers are based on the length scale h, namely $Re=\rho Uh/\mu$ and $Fr=U^2/gh$. [**2 marks**]

In non-dimensional form, the momentum equation is recasted as follows: [**2 marks**]

$$0=-\frac{P}{L}\frac{\partial\tilde{p}}{\partial\tilde{x}}+\frac{\mu U}{h^2}\frac{\partial^2\tilde{u}}{\partial\tilde{y}^2}+\rho g\tilde{g}_x$$

$$0=-\frac{\partial\tilde{p}}{\partial\tilde{x}}+\frac{\mu UL}{Ph^2}\frac{\partial^2\tilde{u}}{\partial\tilde{y}^2}+\frac{\rho gL}{P}\tilde{g}_x$$

By defining the pressure scale $P=\mu UL/h^2$, the first coefficient on the RHS is made equal to unity. [**2 marks**]

With this definition, the second coefficient on the RHS becomes: [**2 marks**]

$$\frac{\rho g L}{P} = \frac{\rho g h^2}{\mu U} = \frac{Re}{Fr}$$

Therefore,

$$0 = -\frac{\partial \tilde{p}}{\partial \tilde{x}} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2} + \frac{Re}{Fr}\tilde{g}_x$$

(c) If pressure depends only on y, $\partial p/\partial x = 0$ in the x-momentum equation. [**1 mark**]

We can integrate the x-momentum equation after defining g_x: [**1 mark**]

$$0 = \mu \frac{d^2 u}{dy^2} + \rho g sin\theta$$

We integrate the equation twice [**2 marks**]:
First integration:

$$\frac{du}{dy} = -\alpha y + C_1$$

Where $\alpha = \rho g sin\theta/\mu$.
Second integration:

$$u = -\frac{\alpha}{2} y^2 + C_1 y + C_2$$

We can now invoke the boundary conditions: [**4 marks**]

- At $y = 0, u = -U$ (no slip boundary condition)
- At $y = h, \tau_{yx} = \mu du/dy = 0$ (stress-free condition)

The first BC yields: $C_2 = -U$
The second BC yields: $C_1 = \alpha h$

Therefore, [**2 marks**]

$$u = \alpha y \left(h - \frac{y}{2}\right) - U$$

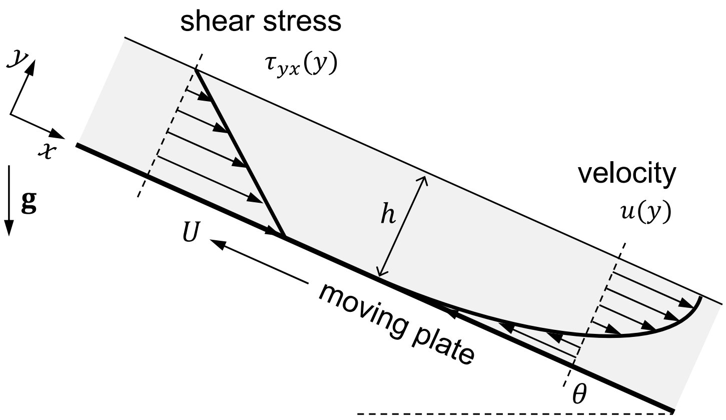
(d) The velocity profile is a quadratic equation, and the velocity profile will therefore be nonlinear, with both positive and negative components. [**2 marks**]
- At $y = 0, u = -U$
- At $y = h, du/dy = 0$ and therefore the velocity will be at its maximum

The shear stress profile has the following equation and is therefore linear:
[**3 marks**]

$$\tau_{yx} = \mu \frac{du}{dy} = \mu\alpha(h - y)$$

- At $y = 0$ the shear stress is at its maximum
- At $y = h, \tau_{yx} = 0$

The following sketch is obtained [**4 marks**]:



QUESTION 2

(a) Natural, free convection

α would need to be calculated from $Nu = \frac{\alpha L}{k}$, where *L* is the characteristic length and *Nu* = *fn* (Re, Pr) and different expressions or graphs could be used for its estimation

[4 marks]

(b) - Assuming sufficient time for all the liquid to reach T_{sat}:

q= $\alpha\ \Delta T_E$ = 9 x 10^3 W m^{-2} K^{-1} x 4 K = 3.6 x 10^4 W m^{-2}

- Assuming maximum sub-cooling is kept:

q= $\alpha\ \Delta T$ = 9 x 10^3 W m^{-2} K^{-1} x (15+4) K = 17.1 x 10^4 W m^{-2}

[4 marks]

(c) - Surface roughness: this affects the onset of nucleate boiling. A smoother surface requires a higher degree of superheat for onset of nucleate boiling; hence the nucleate boiling curve moves to the right as the surface becomes smoother.

- Surface material: this is reflected in C_{sf} in the Rohsenow correlation. For a given excess temperature, the heat flux is proportional to $(C_{sf})^{-3}$, which means that if C_{sf} is doubled, the nucleate boiling heat flux would reduce by 1/8

[4 marks]

(d) - Assuming sufficient time for all the liquid to reach T_{sat}:

Excess temperature ΔT_E = 385 - 375 = 10 K. For water-Pt system, C_{sf}=0.0130 and s=1.0. This is in the nucleate boiling regime. Hence, Rohsenow's correlation for nucleate boiling applies. Data reading from Table 1 is done at T_{sat}=375 K

$$q = \mu_l h_{lv} \left[\frac{g(\rho_l - \rho_v)}{\sigma}\right]^{1/2} \left[\frac{c_{pl}\Delta T_E}{C_{sf} h_{lv}\, Pr_l^s}\right]^3$$

$$= 2.74 \times 10^{-4}(2252 \times 10^3)\left[\frac{9.81(\frac{1}{1.045 \times 10^{-3}} - \frac{1}{1.574})}{0.0586}\right]^{1/2} \left[\frac{4220(10)}{0.013(2252 \times 10^3)(1.70)}\right]^3$$

$$\cong 1.51 \times 10^5\ W\ m^{-2}$$

Total heat transfer rate: $Q = qA = q(\pi DL) = 1.51 \times 10^5\ (3.1415 \times 0.015 \times 0.20) = 1418\ W$

- Assuming maximum sub-cooling is kept, $\Delta T = 10 + 15 = 25\ K$ and liquid properties at T_{sat}, and using Rohsenow's correlation:

$q = 2.35 \times 10^6\ Wm^{-2}$; $Q = qA = q(\pi DL)$=22163 W **[5 marks]**

(e) The minimum heat flux can be calculated by using Zuber's correlation for horizontal cylinder. Assuming sufficient time for all the liquid to reach T_{sat}, so liquid properties are at T_{sat}:

$$q_{\min} = C\rho_v h_{lv}\left[\frac{\sigma g(\rho_l - \rho_v)}{(\rho_l + \rho_v)^2}\right]^{1/4} = 0.09 \times \frac{1}{1.574} \times 2252 \times 10^3 \times \left(\frac{0.0586 \times 9.81\left(\frac{1}{1.045\times10^{-3}} - \frac{1}{1.574}\right)}{\left(\frac{1}{1.045\times10^{-3}} + \frac{1}{1.574}\right)^2}\right)^{1/4} =$$

=2.01 x 10^4 Wm^{-2} A low heat flux compared to previous values despite the high T is due to the presence of a film of gas impairing heat transfer. **[5 marks]**

QUESTION 3

(a) Steam properties are taken at T_{sat} = 365 K

$$\dot{m}' = -\frac{0.004}{\pi D} = -\frac{0.004}{\pi(0.020)} = -0.0637 kgm^{-2}s^{-1}$$

$$Re = \frac{\rho_v U_v D}{\mu_v} = \frac{\frac{1}{2.212}(10)(0.020)}{11.69\times10^{-6}} = 7734.5 \quad ; Pr = \frac{C_{pv}\,\mu_v}{k_v} = \frac{1.999\times10^3(11.69\times10^{-6})}{24.1\times10^{-3}} = 0.970$$

$$Nu = 0.023\,Re^{0.8}\,Pr^{1/3} = 29.38 = \frac{\alpha_o D}{k_v}$$

$$\therefore \alpha_o = \frac{Nuk_v}{D} = \frac{29.38(24.1\times10^{-3})}{0.020} = 35.4\,Wm^{-2}K^{-1}$$

$$\frac{\dot{m}'C_{pv}}{\alpha_o} = -\frac{0.0637(1.999\times10^3)}{35.4} = -3.595$$

$$\frac{\alpha}{\alpha_o} = \frac{\dot{m}'C_{pv}/\alpha_o}{exp(\dot{m}'C_{pv}/\alpha_o)-1} = \frac{-3.595}{exp(-3.595)-1} = 3.696$$

$$\alpha = 3.696\alpha_o = 3.696(35.4) = 130.85\,Wm^{-2}K^{-1}$$

$$\dot{q} = \dot{m}'h_{lv} + \alpha_v(T_{sat}-T_v) = -0.0637(2278\times10^3) - 130.85(10)$$
$$\cong -146.3\times10^3 = -146.3\,kWm^{-2}$$

Assumed laminar film condensation. At this stage, we assumed h_{lv} at saturation temperature but this should be updated because there is large T difference between the interface and the wall as we will see next.

[15 marks]

(b) $\dot{q}' = -U'(T_i - T_c) = -2250\,(365 - T_c) = -146.3\,kWm^{-2}$

$$(T_c) = 365 - \frac{146.3\times10^3}{2250} = 300.0\ K$$

With this value, we can now update h_{lv} and T_c (assuming T_c=T_w)

$$h_{lv}' = h_{lv}\left(1 + 0.68\frac{C_{pl}(T_{sat}-T_w)}{h_{lv}}\right) = 2278\left(1+0.68\left(\frac{(4.209)(365-300)}{2278}\right)\right) =$$

$$= 2278\big(1+0.68(0.120)\big) = 2464\,kJ\,kg^{-1} = 2464\ \times10^3\,J\,kg^{-1}$$

$$\dot{q} = \dot{m}'h_{lv}' + \alpha_v(T_{sat}-T_v) = -0.0637(2464\times10^3) - 130.85(10)$$

$$\cong -158.2\times10^3\,Wm^{-2} = -158.2\,kWm^{-2}$$

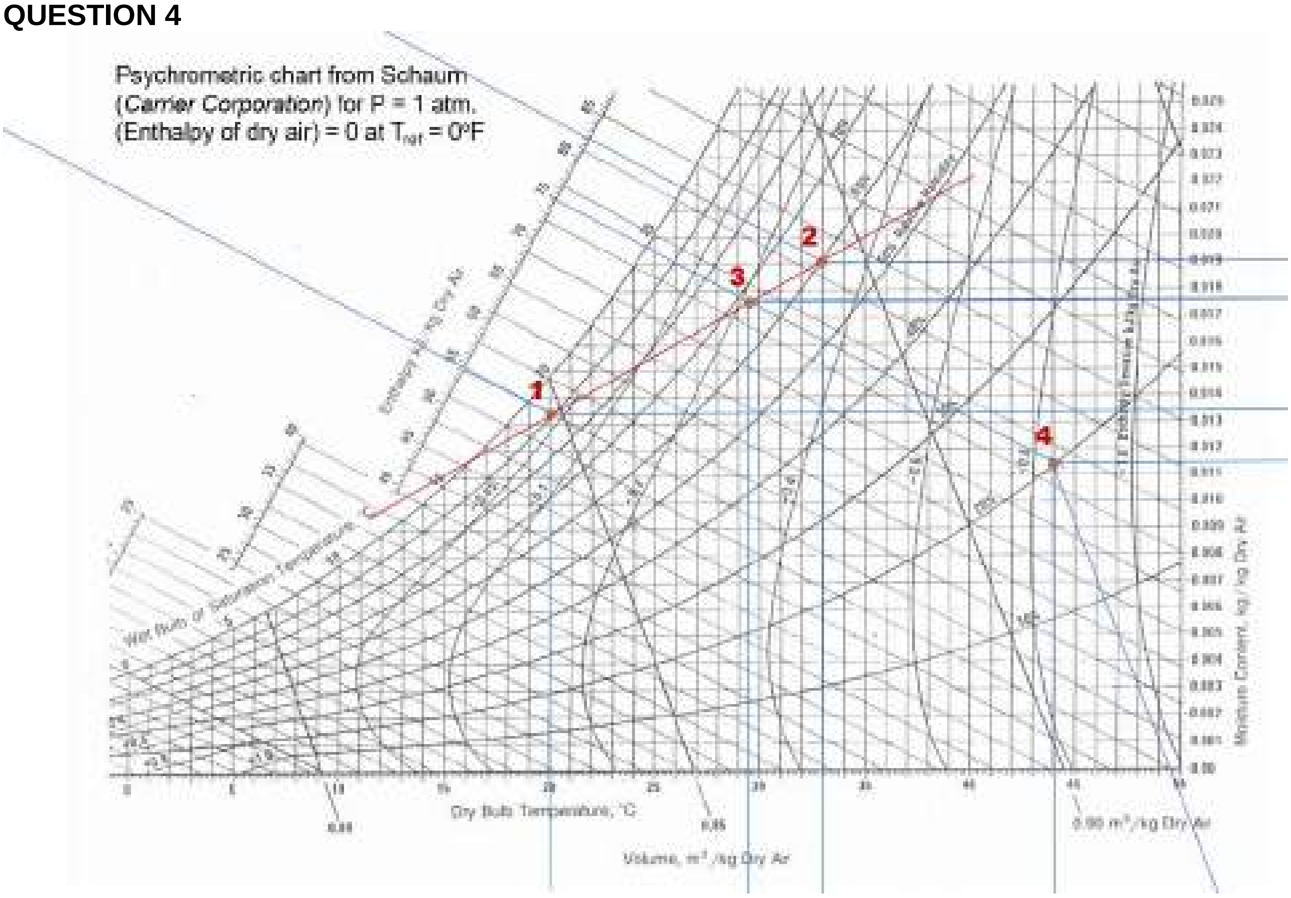
$$T_c = 365 - \frac{-158.1\times10^3}{-2250} = 294.7\ K$$

[5 marks]

(c) Cooling water temperature should be lowered to compensate for the lower interfacial temperature between the gas and the liquid film when non-condensables are present.

[3 marks]

QUESTION 4



(a)

ω_1: 0.0134 kg water (kg dry air)$^{-1}$

T_{wb}: 19.0°C

h_1: 54.4 kJ (kg dry air)$^{-1}$

[5 marks]

(b) Assuming adiabatic

$m_{a1} = \frac{10}{(1+\omega_1)} = \frac{10}{(1+0.0134)} = 9.868$ kg s^{-1}

ω_2: 0.019 kg water (kg dry air)$^{-1}$

$m_{a2} = \frac{30}{(1+\omega_2)} = \frac{30}{(1+0.019)} = 29.441$ kg s^{-1}

$\dot{m}_{a1} + \dot{m}_{a2} = \dot{m}_{a3} = 9.868 + 29.441 = 39.308$ kg s^{-1}

$\dot{m}_{a1}\omega_1 + \dot{m}_{a2}\omega_2 = \dot{m}_{a3}\omega_3 = (\dot{m}_{a1} + \dot{m}_{a2})\omega_3$

$9.868 \times 0.0134 + 29.441 \times 0.019 = 39.308\omega_3$; $\omega_3 = 0.0176$

Reading on graph, T_{db3}=29.5°C

h_3 on graph is 75 kJ (kg dry air)$^{-1}$, which can be confirmed as follows:

$\dot{m}_{a1}h_1 + \dot{m}_{a2}h_2 = \dot{m}_{a3}h_3 = (\dot{m}_{a1} + \dot{m}_{a2})h_3$

$9.868 \times 54.4 + 29.441 \times 82.2 = 39.308\ h_3$

$h_3 = 75.2$ kJ (kg dry air)$^{-1}$

[5 marks]

(c) We assume h_3=h_4 since we do a chemical dehumidification process; therefore, it follows a path of constant specific enthalpy or web-bulb temperature. It is the opposite of “evaporative cooling”. We also assume adiabatic

(i) From graph:

T_{db4}: 44ºC

ν_4: 0.9125 m^3 (kg dry air)$^{-1}$

[5 marks]

(ii) ω_4: 0.0114 kg water (kg dry air)$^{-1}$

The drying rate is $m_{a3}(\omega_4-\omega_3)$= 39.308 (0.0176-0.0114)=0.243 kg s^{-1}

[5 marks]

QUESTION 1

(a) The flow is referred to as Couette flow, because flow is induced by the movement of one boundary tangential to the other. **[2 marks]**

(b) The momentum equation in the z direction reads as follows:

$$\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\left\{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2}\right\} + \mathrm{g}_z$$

[1 mark for each assumption – 7 marks in total]

- Steady-state: $\frac{\partial}{\partial t} = 0$ [drop 1st term LHS]
- Flow is unidirectional: $u_z \neq 0$, $u_r = u_\theta = 0$ [drop: 2nd and 3rd term LHS]
- Flow is fully developed (continuity equation): $\frac{\partial u_z}{\partial z} = 0$ [drop 4th term LHS, 4th term RHS]
- Axisymmetry: $\frac{\partial u_z}{\partial \theta} = 0$
- Pressure gradient is negligible: $\frac{\partial p}{\partial z} = 0$
- Relevant gravity term: g_z
- Partial derivatives can be written as ordinary derivatives

The simplified equations read as follows: **[1 mark]**

$$0 = \frac{\mu}{\rho r}\frac{d}{dr}\left(r\frac{du_z}{dr}\right) + \mathrm{g}_z$$

(c) The stress tensor matrix reads as follows: [**3 marks**]

$$\boldsymbol{\tau} = \begin{bmatrix} 0 & 0 & \mu\frac{du_z}{dr} \\ 0 & 0 & 0 \\ \mu\frac{du_z}{dr} & 0 & 0 \end{bmatrix}$$

(d) The following scalings are used: [**4 marks**]

$$\tilde{r} = \frac{r}{R_2 - R_1} = \frac{r}{h} \qquad \tilde{u}_z = \frac{u_z}{U_0} \qquad \tilde{g}_z = \frac{g_z}{g}$$

In non-dimensional form, the momentum equation is recasted as follows: [**4 marks**]

$$0 = \left[\frac{\mu U_0}{\rho h^2 g}\right]\frac{1}{\tilde{r}}\frac{d}{d\tilde{r}}\left(\tilde{r}\frac{d\tilde{u}_z}{d\tilde{r}}\right) + \tilde{g}_z$$

The Reynolds and Froude numbers are based on the length scale h, namely $Re = \rho U_0 h/\mu$ and $Fr = {U_0}^2/gh$. With this definition [**2 marks**]

$$0 = \frac{Fr}{Re}\frac{1}{\tilde{r}}\frac{d}{d\tilde{r}}\left(\tilde{r}\frac{d\tilde{u}_z}{d\tilde{r}}\right) + \tilde{g}_z$$

(e) We can integrate the z-momentum equation twice [**4 marks**]:

First integration:

$$\frac{d\tilde{u}_z}{d\tilde{r}} = -\frac{1}{2}\frac{Re}{Fr}\tilde{g}_z\tilde{r} + \frac{C_1}{\tilde{r}}$$

Second integration:

$$\tilde{u}_z = -\frac{1}{4}\frac{Re}{Fr}\tilde{g}_z\tilde{r}^2 + C_1\ln(\tilde{r}) + C_2$$

We can now invoke the boundary conditions: [**4 marks**]

- At $r = R_1, u = -U_0$ (no slip boundary condition)
  Using dimensionless variables: at $\tilde{r} = R_1/h\,, \tilde{u} = -1$

- At $r = R_2, u = 0$ (no slip boundary condition)
  Using dimensionless variables: at $\tilde{r} = R_2/h\,, \tilde{u} = 0$

The BCs yield: [**2 marks**]

$$C_1 = \frac{1 + \frac{1}{4}\tilde{g}_z\left(\frac{Re}{Fr}\right)\left[\left(\frac{R_2}{h}\right)^2 - \left(\frac{R_1}{h}\right)^2\right]}{\ln\left(R_2/R_1\right)}$$

$$C_2 = \frac{\ln\left(R_2/h\right)}{\ln\left(R_2/R_1\right)}\left\{\frac{1}{4}\tilde{g}_z\left(\frac{Re}{Fr}\right)\left[\left(\frac{R_1}{h}\right)^2 + \left(\frac{R_2}{h}\right)^2\left(\frac{\ln\left(R_2/R_1\right)}{\ln\left(R_2/h\right)} - 1\right) - 1\right]\right\}$$

(f) The solutions in terms of non-dimensional variables (*e.g.* $\tilde{u}$) are identical for two systems with equivalent Re and Fr. [**2 marks**]

QUESTION 2

(a) In the film boiling regime, the heated surface is covered by a vapour film, so that heat is transferred by conduction through the vapour film and then convection at the vapour/liquid interface. Since vapour is generally a poor conductor and heat transfer by radiation is insignificant at relative low surface temperature, the total heat transfer coefficient is low near the minimum heat flux point. At high surface temperature, radiative heat transfer becomes significant (since heat flux is proportional to ($T_w^4 - T_{sat}^4$) thereby increasing the total heat transfer coefficient.

[5 marks]

(b) The minimum heat flux can be calculated by using Zuber's correlation for horizontal plates.

$$q_{\min} = C\rho_v h_{lv}\left[\frac{\sigma g(\rho_l-\rho_v)}{(\rho_l+\rho_v)^2}\right]^{1/4} = 0.09 \times \frac{1}{1.142} \times 2225 \times 10^3 \times \left(\frac{0.0566\times 9.81\left(\frac{1}{1.053\times 10^{-3}}-\frac{1}{1.142}\right)}{\left(\frac{1}{1.053\times 10^{-3}}+\frac{1}{1.142}\right)^2}\right)^{1/4} =$$

$=2.72 \times 10^4\, Wm^{-2}$ **[5 marks]**

(c) The ideal two-phase flow regime for heat transfer is annular flow, where liquid film moves along the inner surface, while vapour moves at a higher velocity through the core of the tube. In this regime, nucleate boiling is partially suppressed, and heat transfer is dominated by forced convection through liquid film. The heat transfer coefficient is very high due to the high velocity of gas and the evaporation at the gas-liquid interface.

[5 marks]

(b) Rohsenow used the following expression for *L*, which was taken from Taylor wave theory:

$$L = \sqrt{\frac{\sigma}{g(\rho_l-\rho_v)}}$$

and, this is used in the Nu number as follows:

$$Nu = \frac{\alpha L}{k_l} = \frac{\alpha}{k_l}\sqrt{\frac{\sigma}{g(\rho_l-\rho_v)}}$$

For the average velocity used in Re, he considered that bubbles behave as micropumps, so the liquid velocity towards the heated surface in order to fill behind a departing bubble will be the following:

$$U = \frac{\dot{m}_l}{\rho_l A} = \frac{\dot{m}_v}{\rho_l A} = \frac{Q}{\rho_l A h_{lv}} = \frac{q}{\rho_l h_{lv}}$$

As a result, Re can be expressed as:

$$Re = \frac{\rho_l U L}{\mu_l} = \frac{\rho_l}{\mu_l}\frac{q}{\rho_l h_{lv}}\sqrt{\frac{\sigma}{g(\rho_l-\rho_v)}} = \frac{q}{\mu_l h_{lv}}\sqrt{\frac{\sigma}{g(\rho_l-\rho_v)}}$$

[7 marks]

QUESTION 3

(a)

Using the velocity profile equation provided

$$\Gamma(x) = \frac{\dot{m}(x)}{w} = \int_0^\delta \rho_l u(y) dy = \frac{\rho_l(\rho_l - \rho_v)g\delta^2}{\mu_l} \int_0^\delta \left[\frac{y}{\delta} - \frac{1}{2}\left(\frac{y}{\delta}\right)^2\right] dy = \frac{\rho_l(\rho_l - \rho_v)g\delta^3}{3\mu_l}$$

[5 marks]

(b) At a portion of the liquid-vapour interface of length, dx, the rate of heat transfer into the film, dQ, must equal the rate of energy release due to condensation at the interface:

$$dQ = h_{lv} d\dot{m}$$

Since convection is neglected, it also follows that the rate of heat transfer across the interface must equal the rate of heat transfer across the surface. Hence $dQ = q(w \cdot dx)$ which gives $q = \dfrac{dQ}{wdx} = \dfrac{h_{lv} d\dot{m}}{wdx} = h_{lv} \dfrac{d\Gamma}{dx}$.

q is given from Fourier's Law: $q = k_l \dfrac{T_{sat} - T_w}{\delta}$, since the liquid temperature distribution is linear. Hence,

$$h_{lv} \frac{d\Gamma}{dx} = k_l \frac{T_{sat} - T_w}{\delta}$$

or,

$$\frac{d\Gamma}{dx} = \frac{k_l (T_{sat} - T_w)}{\delta h_{lv}}$$

Using the expression for Γ derived from the momentum equation, we can obtain:

$$\frac{d\Gamma}{dx} = \frac{\rho_l(\rho_l - \rho_v)g\delta^2}{\mu_l} \frac{d\delta}{dx}$$

Combining the above equations, we have

$$\delta^3 d\delta = \frac{k_l \mu_l (T_{sat} - T_w)}{\rho_l(\rho_l - \rho_v)g h_{lv}} dx$$

which can be integrated once subject to the boundary condition, $\delta(x = 0) = 0$.
The result is:

$$\delta(x) = \left[\frac{4k_l \mu_l (T_{sat} - T_w)x}{\rho_l(\rho_l - \rho_v)g h_{lv}}\right]^{1/4}$$

[10 marks]

(c) Finally, the heat transfer coefficient can be obtained from

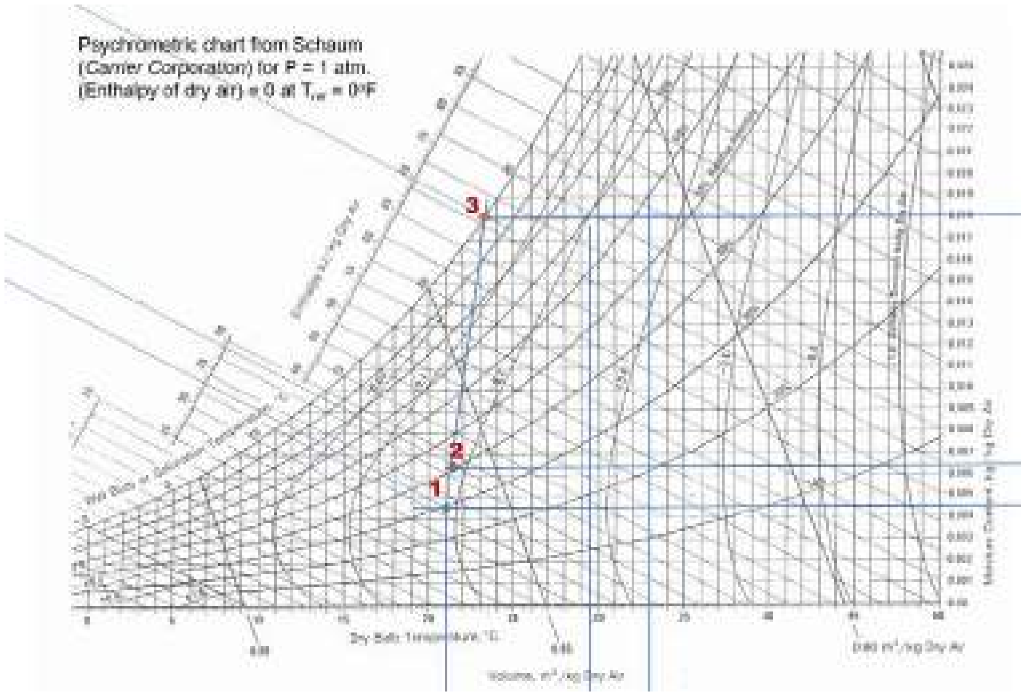
$$\alpha = \frac{q}{T_{sat} - T_w} = \frac{1}{T_{sat} - T_w}\left(\frac{k_l (T_{sat} - T_w)}{\delta}\right) = \frac{k_l}{\delta} = \left[\frac{\rho_l(\rho_l - \rho_v)g k_l^3 h_{lv}}{4\mu_l (T_{sat} - T_w)x}\right]^{1/4}$$

Since α depends on $x^{-1/4}$, it follows that the average heat transfer coefficient over the entire plate length L is:

$$\overline{\alpha} = \frac{1}{L}\int_0^L \alpha dx = 0.943\left[\frac{\rho_l(\rho_l - \rho_v)g k_l^3 h_{lv}}{\mu_l (T_{sat} - T_w)L}\right]^{1/4}$$

[5 marks]

QUESTION 4



(a) RH$_1$: 30%

T_{wb}: 11.5ºC

h_1: 33 kJ (kg dry air)$^{-1}$

[5 marks]

(b) Assuming adiabatic

(i)

$$\dot{m}_{a1} = \dot{m}_{a2}$$

$$\dot{m}_{a1}h_1 + \dot{m}_w h_w = \dot{m}_{a2}h_2$$

$$\dot{m}_w h_w = \dot{m}_{a1}(h_2 - h_1)$$

$$0.020 \times 2676 = 10(h_2 - 33)$$

$$h_2 = \frac{0.020\times 2676}{10} + 33 = 38.352 \text{ kJ (kg dry air)}^{-1}$$

[5 marks]

(ii)

$$\dot{m}_{a1}\omega_1 + \dot{m}_w = \dot{m}_{a2}\omega_2$$

$$\omega_2 = \frac{\dot{m}_{a1}\omega_1 + \dot{m}_w}{\dot{m}_{a1}} = \frac{10\ \times 0.0045 + 0.020}{10} = 0.0065$$

From graph: RH$_2$: 40%

[5 marks]

(iii) Using graph, extrapolating to when RH:100 %, ω_3=0.018 kg kg^{-1}(dry air)

$$\dot{m}_{a1}\omega_1 + \dot{m}_w = \dot{m}_{a3}\omega_3$$

$$\dot{m}_w = \dot{m}_{a1}\omega_3 - \dot{m}_{a1}\omega_1 = 10(0.018 - 0.0045) = 0.135 \text{ kg s}^{-1}$$

[5 marks]

(c) Doing a chemical dehumidification process; for example, passing the stream through a desiccant silica gel

[3 marks]

QUESTION 1

(a) The lubrication approximation applies to flow in a narrow gap. This flow possesses two characteristic length scales (gap height, a, and length of passage, L), such that:

$$\varepsilon = \frac{a}{L} \ll 1$$

(b) The following scalings are used:

$$\tilde{x} = \frac{x}{L} \qquad \tilde{y} = \frac{y}{a} = \frac{y}{\varepsilon L} \qquad \tilde{u} = \frac{u}{u_b} \qquad \tilde{v} = \frac{v}{V}$$

The characteristic velocity scale, V, is obtained by making the continuity equation dimensionless, i.e.

$$\frac{\partial \tilde{u}}{\partial \tilde{x}} + \frac{\partial \tilde{v}}{\partial \tilde{y}} = 0 \qquad \text{when} \quad V = \frac{u_b a}{L} = \varepsilon u_b$$

(c) The momentum equation in the x direction reads as follows:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\left\{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right\} + g_x$$

[

- Steady-state: $\frac{\partial}{\partial t} = 0$ [drop 1st term LHS]
- Gravity can be neglected, $g_x = 0$ [drop 4th term RHS]

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{dp}{dx} + \nu\left\{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right\}$$

Upon substitution of all dimensionless variables, the following expression is obtained:

$$\left[\frac{\rho u_b^2}{L}\right]\left(\tilde{u}\frac{\partial \tilde{u}}{\partial \tilde{x}} + \tilde{v}\frac{\partial \tilde{u}}{\partial \tilde{y}}\right) = -\left[\frac{P}{L}\right]\frac{d\tilde{p}}{d\tilde{x}} + \left[\frac{\mu u_b}{\varepsilon^2 L^2}\right]\left(\varepsilon^2\frac{\partial^2 \tilde{u}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2}\right)$$

We divide the equation by $[\mu u_b/\varepsilon^2 L^2]$ and substitute $Re_L = \rho u_b L/\mu$ to obtain:

$$[\varepsilon^2 Re_L]\left(\tilde{u}\frac{\partial \tilde{u}}{\partial \tilde{x}} + \tilde{v}\frac{\partial \tilde{u}}{\partial \tilde{y}}\right) = -\left[\frac{a^2 P}{L\mu u_b}\right]\frac{d\tilde{p}}{d\tilde{x}} + \left(\varepsilon^2\frac{\partial^2 \tilde{u}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2}\right)$$

The final, simplified equation is obtained upon defining the pressure scale as $P = L\mu u_b/a^2$ and cancelling terms multiplied by ε^2 (note that $Re = \rho u_b L/\mu \approx 1$).

$$0 = -\frac{d\tilde{p}}{d\tilde{x}} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2}$$

(d) We can integrate the x-momentum equation twice

First integration:

$$\frac{d\tilde{u}}{d\tilde{y}} = \frac{d\tilde{p}}{d\tilde{x}}\tilde{y} + C_1$$

Second integration:

$$\tilde{u} = \frac{1}{2}\frac{d\tilde{p}}{d\tilde{x}}\tilde{y}^2 + C_1\tilde{y} + C_2$$

We can now invoke the boundary conditions:

- At $\tilde{y} = 0, \tilde{u} = 0$ (no slip boundary condition), yielding $C_2 = 0$
- At $\tilde{y} = 1, \tilde{u} = 1$ (no slip boundary condition), yielding $C_1 = 1 - \frac{1}{2}\frac{d\tilde{p}}{d\tilde{x}}$

$$\tilde{u} = \frac{1}{2}\left(-\frac{d\tilde{p}}{d\tilde{x}}\right)(\tilde{y} - \tilde{y}^2) + \tilde{y}$$

This equation is then recasted in dimensional form by using the definition of the dimensionless variables to obtain:

$$u = \frac{1}{\mu}\left(-\frac{dp}{dx}\right)\left[\frac{1}{2}(ay - y^2) + \frac{yk}{a}\right]$$

Note that as given in the problem statement,

$$u_b = -\frac{k}{\mu}\frac{dp}{dx}$$

(e) The only relevant component of the stress tensor is $\tau_{yx} = \mu\, du/dy$.

Therefore, we can differentiate the velocity profile given in question (d) to obtain:

$$\frac{du}{dy} = \frac{1}{\mu}\left(-\frac{dp}{dx}\right)\left[\frac{1}{2}(a - 2y) + \frac{k}{a}\right]$$

Finally,

$$\tau_{yx}(y = a) = \left(-\frac{dp}{dx}\right)\left[-\frac{a}{2} + \frac{k}{a}\right]$$

QUESTION 2

(a)

(i) Circles, because according to the equation, $R_b = \frac{2\sigma T_{sat}}{h_{lv}\Delta T \rho_v}$, excess of temperature decreases with bubble radius; therefore, a rougher surface (and larger nucleated bubbles) will happen at smaller excess of temperature.

(ii) 4 regimes:

Natural convection boiling (not present in Figure 2): No bubble formation, heat transfer by natural convection.

Nucleate boiling: Bubbles form by nucleation at the surface, bubbles interfere and coalesce

Transitional boiling: Rapid bubble formation, vapour film on the surface, unstable regime

Film boiling: Surface completely covered by vapour film, heat transfer by conduction through the vapour film, radiation significant when surface becomes very hot.

The only one affected by surface conditions is the nucleate boiling regime. The others are only dependent on physical properties.

(b)

(i) Data reading for Question from Table 1:

Liquid Density	ρ_l	1000/1.041=960.6 kg m^{-3}
Vapour Density	ρ_v	1/1.861=0.537 kg m^{-3}
Latent heat of vaporisation	h_{lv}	2265 kJ kg^{-1}
Surface tension	σ	0.0595 N m^{-1}

$$q_{max} = 0.131\rho_v^{\frac{1}{2}} \; h_{lv}[\sigma g(\rho_l - \rho_v)]^{\frac{1}{4}}$$

$$= 0.131 \times 0.537^{\frac{1}{2}} \times 2265000\left(0.0595 \times 9.81 \times (960.6 - 0.537)\right)^{\frac{1}{4}} = 1057908\, Wm^{-2}$$

Q=qA ; A=Q/q=500000 W / (1057908 Wm^{-2})= 0.22 m^2 minimal area

(ii) Rearranging Rohsenow equation, we can calculate the excess of temperature above saturation temperature as follows:

$$q = \mu_l h_{lv}\left[\frac{g(\rho_l - \rho_v)}{\sigma}\right]^{1/2}\left[\frac{c_{pl}\Delta T_E}{C_{sf} h_{lv}\, Pr_l^s}\right]^3$$

$$\Delta T_E = \left(\frac{q}{\mu_l h_{lv}\left[\frac{g(\rho_l - \rho_v)}{\sigma}\right]^{\frac{1}{2}}}\right)^{\frac{1}{3}} \times \frac{C_{sf} h_{lv}\, Pr_l^s}{c_{pl}}$$

$$= \left(\frac{1057908}{2.89 \times 10^{-4}(2265 \times 10^{3}) \left[\frac{9.81\left(\frac{1}{1.041 \times 10^{-3}} - \frac{1}{1.861}\right)}{0.0595} \right]^{1/2}} \right)^{1/3} \frac{0.013(2265 \times 10^{3})(1.80)}{4214}$$

$$= 20.07\ \mathrm{K}$$

Assuming $T_{wall} = T_{steam}$

$T_{steam} = T_{sat} + \Delta T_E = 370 + 20.07 = 390.1$ K, which corresponds to 1.8 bar of maximum P_{steam}

(iii) $\dot{m} = \frac{Q}{h_{lv}} = \frac{500000}{2265000} = 0.22$ kg s^{-1}

QUESTION 3

(a)

(i) The momentum equation for the condensate film is:

$$\rho_l \left(\cancel{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}} \right) = -\cancel{\frac{\partial p}{\partial x}} + \mu_l \frac{\partial^2 u}{\partial y^2} + \cancel{\rho_l g}$$

So the approximate momentum equation becomes:

$$\mu_l \frac{\partial^2 u}{\partial y^2} = 0$$

Boundary conditions for the above equation are:

$u(y = 0) = 0$; no-slip condition

$\mu_l \left. \frac{\partial u}{\partial y} \right|_{y=\delta} = \tau_i$; assuming shear stress at the liquid-vapour interface is known.

Integrating the approximate momentum equation twice subject to the BCs, the velocity profile in the condensate film can be obtained.

$$u(y) = \frac{\tau_i}{\mu_l} y$$

This is a linear profile and the mean velocity in the condensate film is given by

$$\bar{u} = \frac{\tau_i}{2\mu_l} \delta$$

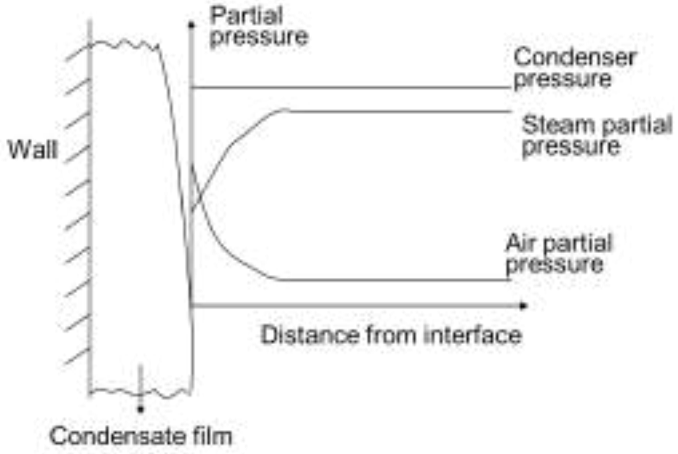
<u>(ii)</u> Mass (or condensate) flow rate per unit width is:

$$\Gamma = \rho_l \bar{u} \delta = \frac{\rho_l \delta^2 \tau_i}{2\mu_l}$$

$$\delta = \sqrt{\frac{2\Gamma \mu_l}{\tau_i \rho_l}}$$

$$\alpha = \frac{k_l}{\delta} = \sqrt{\frac{k_l^2 \tau_i \rho_l}{2\Gamma \mu_l}}$$

(b)
(i)



(ii)

$$y_{1v} = \frac{0.2/18.02}{0.8/28.96 + 0.2/18.02} = 0.2866$$

$$C_T = \frac{p}{RT} = \frac{1.5 \times 10^5}{8.314(273 + 180)} = 39.8\, molm^{-3}$$

$$\dot{m}_1^{'} = -\frac{0.002}{\pi D} = -\frac{0.002}{\pi 0.05} = -0.0127 kgm^{-2}s^{-1}$$

$$\dot{n}_1 = \frac{\dot{m}_1'}{M_{water}} = \frac{-0.0127 \times 1000}{18.08} = -0.707\, molm^{-2}s^{-1}$$

$$Re = \frac{\rho_v u_v D}{\mu_v} = \frac{0.744(30)(0.05)}{2.57 \times 10^{-5}} = 43424$$

$$Sc = \frac{\mu_v}{\rho_v D_{12}} = \frac{2.57 \times 10^{-5}}{0.744 \ \times 5.12 \ \times 10^{-5}} = 0.675$$

$$Sh = 0.023 Re^{0.8} Sc^{1/3} = 0.023 \ \times 43424^{0.8} 0.675^{1/3} = 103.5 = \frac{\beta D}{D_{12}}$$

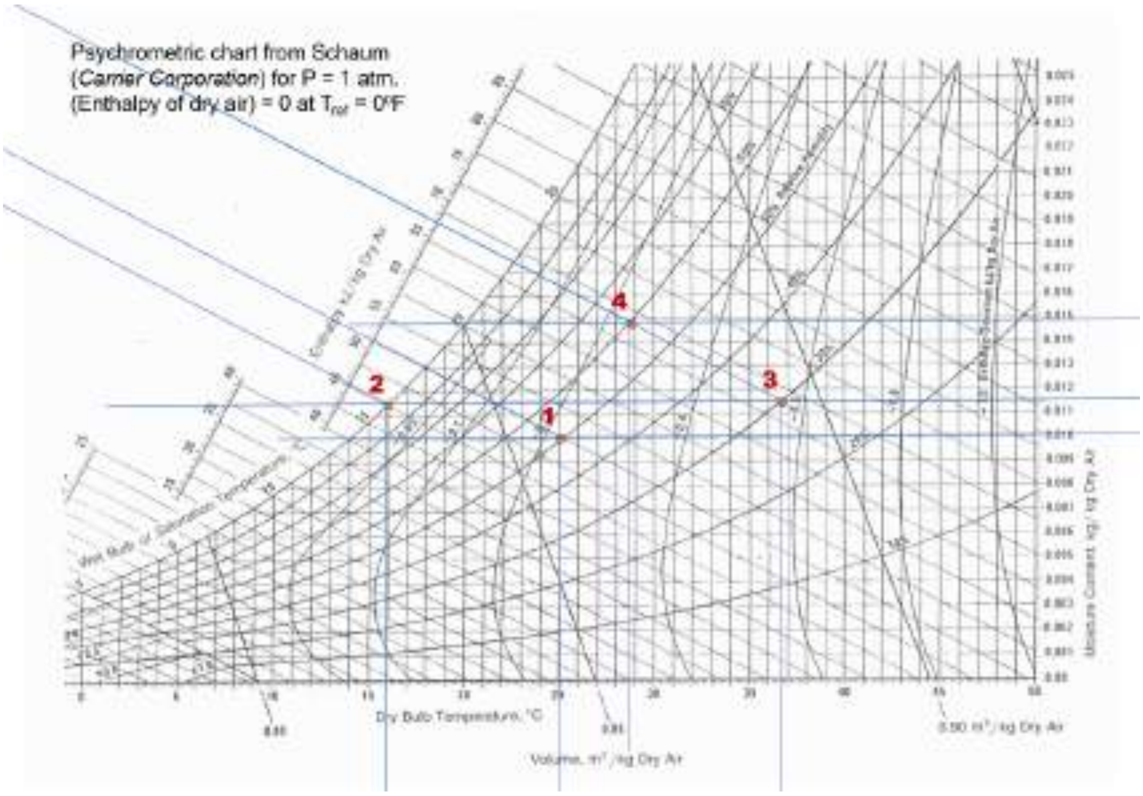
$$\therefore \beta = \frac{Sh D_{12}}{D} = \frac{103.5 \ \ 5.12 \times 10^{-5}}{0.05} = 0.106\, ms^{-1}$$

$$\dot{n}_1 = \beta C_T \, ln\left[\frac{1 - y_{1v}}{1 - y_{1i}}\right] =$$

$$\therefore -0.707 = 0.106(39.8)\, ln\left[\frac{1 - 0.2866}{1 - y_{1i}}\right]$$

$y_{1i} = 0.1565$; which is below 0.2866, it makes sense
$p_{1i} = y_{1i} p = 0.1565 \times 1.5 = 0.235\ bar$
From the steam table, you can find that $T_i \approx$ 336.7K.

QUESTION 4



(a) Using the psychrometric chart:

ω_1= 0.010 kg water (kg dry air)$^{-1}$

T_{wb}= 18ºC

T_{DP}= 14.2 ºC

(b)

i. T_{db}= 36.6 ºC

ii. Heat provided is 67-45=22 kJ (kg dry air)$^{-1}$

To calculate the (kg dry air) h^{-1}, we do:

ω_2= 0.0116 (kg water) (kg dry air)$^{-1}$

$m_{a2} = \frac{10}{(1+\omega_1)} = \frac{10}{(1+0.0116)} = 9.885$ (kg dry air) h^{-1}

then the heat provided per hour is

9.885 (kg dry air) h^{-1} x 22 kJ (kg dry air)$^{-1}$ = 217.5 kJ h^{-1}

iii. The condensate is ω_4-ω_3= 0.0148-0.0116=0.0032 kg water (kg dry air)$^{-1}$

9.885 (kg dry air) h^{-1} x 0.0032 (kg water) (kg dry air)$^{-1}$= 0.0316 kg water h^{-1}

2025
QUESTION 1

(a) To verify that $p = p(x)$, we must demonstrate that pressure does not depend on y. To this end, we first consider the continuity equation, formulated in 2D: [2 marks]

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

The flow is fully developed, $\frac{\partial}{\partial x} \to 0$, and therefore $\frac{\partial v}{\partial y} = 0$. We shall now consider and simplify the y-component momentum equation [3 marks]:

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right] + g_y$$

- Steady-state: $\frac{\partial v}{\partial t} = 0$ [drop 1st term RHS]
- From the continuity equation (above): $\frac{\partial}{\partial x} \to 0$ and $\frac{\partial v}{\partial y} = 0$ [drop 2nd, 3rd LHS and RHS].
- Gravity is considered unimportant,: $g_y = 0$ [drop 4th term RHS].

The simplified equation reads:

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

i.e. pressure does not depend on y. [1 mark]

(b) x-component momentum equation: [4 marks]

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right] + g_x$$

- Steady-state: $\frac{\partial u}{\partial t} = 0$ [drop 1st term LHS]
- Fully developed flow: $\frac{\partial}{\partial x} \to 0$; thus, $\frac{\partial u}{\partial x} = 0$ and $\frac{\partial^2 u}{\partial x^2} = 0$ [drop 2nd term LHS and RHS].
- The flow is unidirectional: $v = 0$ [drop 3rd LHS].
- Gravity is considered unimportant,: $g_x = 0$ [drop 4th term RHS].

The simplified equation reads as follows (in each layer): [1 mark]

$$0 = -\frac{dp}{dx} + \mu\frac{d^2 u}{dy^2}$$

(c) Upon integration the following expressions are obtained:

1st integration: [2 marks]

$$\frac{du_A}{dy} = \frac{1}{\mu_A}\frac{dp}{dx}y + c_{1A}$$

2nd integration: [2 marks]

$$u_A = \frac{1}{2\mu_A}\frac{dp}{dx}y^2 + c_{1A}y + c_{2A}$$

(d) We must solve for the following two equations:

$$u_A = \frac{1}{2\mu_A}\frac{dp}{dx}y^2 + c_{1A}y + c_{2A}$$

$$u_B = \frac{1}{2\mu_B}\frac{dp}{dx}y^2 + c_{1B}y + c_{2B}$$

The following four boundary conditions apply: [4 marks]

1. No-slip condition at bottom plate: at $y = 0$, $u_A = 0$
2. No-slip condition at upper plate: at $y = H$, $u_B = 0$
3. No-slip condition at interface: at $y = d$, $u_A = u_B$
4. Continuity of stress at interface: at $y = d$, $\tau_{xy}\big|_{\mu_A, y=d} = \tau_{xy}\big|_{\mu_B, y=d}$

[1 mark for each expression] = [4 marks]

- From (4):

$\tau_{xy}\big|_{\mu_A, y=d} = \tau_{xy}\big|_{\mu_B, y=d}$, where $\tau_{xy}\big|_{\mu, y=d} = \mu \frac{du}{dy}\big|_{y=d}$

$$\mu_A\left(\frac{1}{\mu_A}\frac{dp}{dx}d + c_{1A}\right) = \mu_B\left(\frac{1}{\mu_B}\frac{dp}{dx}d + c_{1B}\right) \rightarrow c_{1A} = \frac{\mu_B}{\mu_A}c_{1B}$$

- From (1):

$$c_{2A} = 0$$

- From (2):

$$c_{2B} = -\frac{1}{2\mu_B}\frac{dp}{dx}H^2 - c_{1B}H$$

- From (3) and upon combining the above expressions:

$$\frac{1}{2\mu_A}\frac{dp}{dx}d^2 + c_{1A}d = \frac{1}{2\mu_B}\frac{dp}{dx}d^2 + c_{1B}d + c_{2B}$$

$$\frac{1}{2\mu_A}\frac{dp}{dx}d^2 + \frac{\mu_B}{\mu_A}c_{1B}d = \frac{1}{2\mu_B}\frac{dp}{dx}d^2 + c_{1B}d - \frac{1}{2\mu_B}\frac{dp}{dx}H^2 - c_{1B}H$$

$$c_{1B} = -\frac{1}{2\mu_B}\frac{dp}{dx}H\frac{\left(\frac{d}{H}\right)^2\left(1 - \frac{\mu_B}{\mu_A}\right) - 1}{\left(\frac{d}{H}\right)\left(1 - \frac{\mu_B}{\mu_A}\right) - 1}$$

It follows, that

$$c_{1A} = -\frac{1}{2\mu_A}\frac{dp}{dx}H\frac{\left(\frac{d}{H}\right)^2\left(1-\frac{\mu_B}{\mu_A}\right)-1}{\left(\frac{d}{H}\right)\left(1-\frac{\mu_B}{\mu_A}\right)-1}$$

$$c_{2B} = -\frac{1}{2\mu_B}\frac{dp}{dx}H^2\left(1-\frac{\left(\frac{d}{H}\right)^2\left(1-\frac{\mu_B}{\mu_A}\right)-1}{\left(\frac{d}{H}\right)\left(1-\frac{\mu_B}{\mu_A}\right)-1}\right)$$

The velocity profile for liquid A is: [1 mark]

$$u_A = \frac{1}{2\mu_A}\frac{dp}{dx}H^2\left[\left(\frac{y}{H}\right)^2-\left(\frac{y}{H}\right)\left(\frac{\left(\frac{d}{H}\right)^2\left(1-\frac{\mu_B}{\mu_A}\right)-1}{\left(\frac{d}{H}\right)\left(1-\frac{\mu_B}{\mu_A}\right)-1}\right)\right]$$

The velocity profile for liquid B is: [1 mark]

$$u_B = \frac{1}{2\mu_B}\frac{dp}{dx}H^2\left[\left(\frac{y}{H}\right)^2-\left(\frac{\left(\frac{d}{H}\right)^2\left(1-\frac{\mu_B}{\mu_A}\right)-1}{\left(\frac{d}{H}\right)\left(1-\frac{\mu_B}{\mu_A}\right)-1}\right)\left(\left(\frac{y}{H}\right)-1\right)-1\right]$$

(e) The stress profile can now be readily obtained: [2 marks]

$$\tau_{xy} = \mu\frac{du}{dy}$$

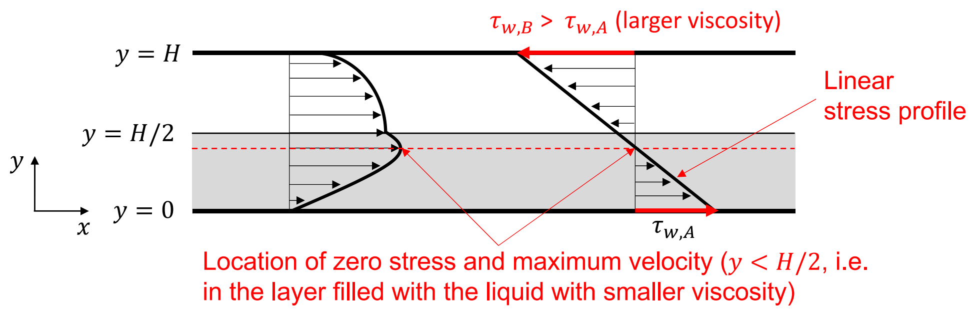
We note that there is only one expression for the stress profile, because [1 mark]

$$\mu_A\frac{du_A}{dy} = \mu_B\frac{du_B}{dy}$$

As such [2 marks]

$$\tau_{xy} = \frac{dp}{dx}\left[y-\frac{H}{2}\left(\frac{\left(\frac{d}{H}\right)^2\left(1-\frac{\mu_B}{\mu_A}\right)-1}{\left(\frac{d}{H}\right)\left(1-\frac{\mu_B}{\mu_A}\right)-1}\right)\right]$$

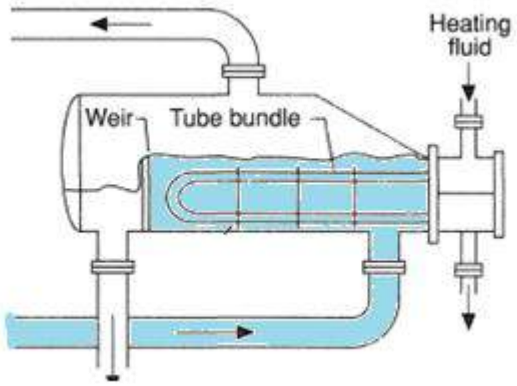
(d) [2 marks for sketch and 3 marks for indicating the three features] = [5 marks]



QUESTION 2 **[24 points]**

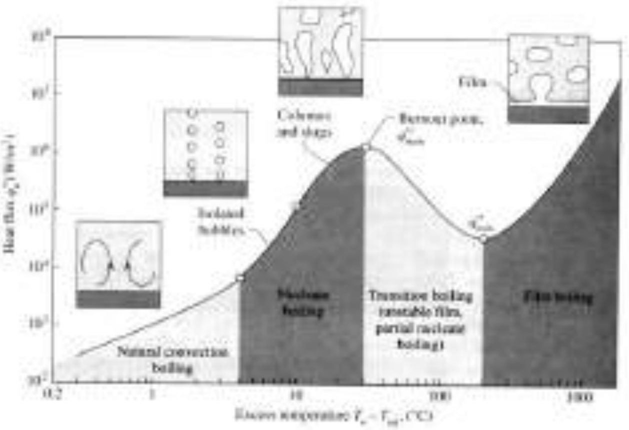
(a)

(i) Pool boiling occurs because the liquid is contained by the weir, like in a pool. There is little motion, as drawn below:



[3 marks]

(ii) Sketch of boiling curve:



In natural convection boiling, the system remains in a single phase and heat is transferred by convection in the liquid. In nucleate boiling, heat transfer is promoted by the rise of bubbles, which causes forced convection in the liquid. In the film boiling regime, the heated surface is covered by a vapour film, so that heat is transferred by conduction through the vapour film and then convection at the vapour/liquid interface. For sufficiently high temperature (T_w > 300 C) there is also a contribution from radiation across the vapour film

[3 marks]

(iii) The burnout point should be avoided in the design of boilers because near this point a small increase in the heat flux (in power-controlled mode) can lead to a very large increase in temperature of the heater surface. The temperature can exceed the melting temperature of the metal, causing damage to equipment. This can be prevented if the equipment is operated in temperature-controlled mode.

[3 marks]

(b)

(i) Excess temperature ΔT_E = 380-100 = 280°C. This is in the film boiling regime. Heat flux now has two components, convective and radiative. $q_{total} = \alpha_{total} \Delta T_E$

Properties for water vapour in film estimated at mean film temperature (380+100)/2=240°C=513K. From the property table we have:

ρ_v=0.4295 kgm^{-3}	v_v = 41.47×10^{-6} m^2s^{-1}	ρ_l=958 kgm^{-3}
Cp_v = 1986 $Jkg^{-1}K^{-1}$	k_v=0.03716 $Wm^{-1}K^{-1}$	h_{lv}=2257 $kJkg^{-1}$

$$h'_{lv} = h_{lv} + 0.80Cp_v(T_W - T_{sat}) = 2257000 + 0.80(1986)(280) = 2702\ kJkg^{-1}$$

$$Nu_D = C\left[\frac{g(\rho_l - \rho_v)h'_{lv}D^3}{\nu_v k_v(T_W - T_{sat})}\right]^{1/4}$$

where C=0.62 for horizontal cylinders and D=0.04 m, so that

$$Nu_D = \frac{\alpha_{conv}D}{k_v} = 0.62\left[\frac{9.81(958-0.4295)2702\times10^3(0.04)^3}{41.47\times10^{-6}(0.03716)(280)}\right]^{1/4} = 153.6$$

$$\alpha_{conv} = \frac{Nuk_v}{D} = \frac{(153.6)(0.03716)}{0.04} = 142.7\ Wm^{-2}K^{-1}$$

$$\alpha_{rad} = \frac{\varepsilon\sigma(T_W^4 - T_{sat}^4)}{T_W - T_{sat}} = \frac{0.8\ (5.67\times10^{-8})(653^4 - 373^4)}{653 - 373} = 26.3\ Wm^{-2}K^{-1}$$

$$\alpha_{average} = (\alpha_{conv} + \alpha_{rad})/2 = 90.0\ Wm^{-2}K^{-1}$$

[10 marks]

Since $\alpha_{rad} < \alpha_{conv}$

$$\alpha_{total} = \alpha_{conv} + \frac{3}{4}\alpha_{rad} = 162.4\ Wm^{-2}K^{-1}$$

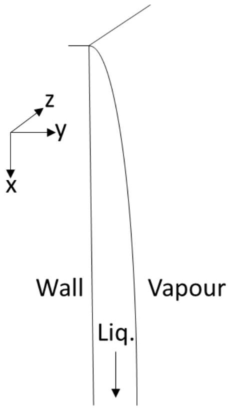
$$q_{total} = 162.4\ (280) = 45481\ \ Wm^{-2}$$

$$Q = qA = q(N\pi DL) = 45481\ (20\ \pi\ \ 0.04\ \ 5) = \ 571521\ W$$

Nickel has lower surface emissivity than the steel tube surface (0.35 vs 0.8), so α_{rad} would be even smaller, lowering the radiation heat transfer further and total heat transfer rate.

[5 marks]

QUESTION 3 **[24 points]**



(a)

$$\cancel{\frac{\partial u}{\partial t}} + u\cancel{\frac{\partial u}{\partial x}} + \cancel{v\frac{\partial u}{\partial y}} + \cancel{w\frac{\partial u}{\partial z}} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \frac{\partial^2 u}{\partial y^2} + \cancel{\frac{\partial^2 u}{\partial z^2}}\right) + g_x$$

Assumptions: steady state, no advection, laminar flow, no interfacial shear….

$$0 = -\frac{\partial p}{\partial x} + \mu_l \frac{\partial^2 u}{\partial y^2} + \rho_l g$$

where the pressure gradient can be approximated in terms of conditions outside the film,

$$\frac{\partial p}{\partial x} \cong \frac{dp_\infty}{dx} = \rho_v g$$

$$0 = (\rho_l - \rho_v)g + \mu_l \frac{\partial^2 u}{\partial y^2}$$

The approximate momentum equation then becomes:

$$\frac{\partial^2 u}{\partial y^2} = -\frac{(\rho_l - \rho_v)g}{\mu_l}$$

[8 marks]

(b) Boundary conditions for the above equations are:

$u(y=0)=0$; non-slip condition

$\left.\frac{\partial u}{\partial y}\right|_{y=\delta} = 0$; zero shear stress at the liquid-vapour interface

Integrating the approximate momentum equation twice subject to the BCs, the velocity profile in the film becomes:

$$u(y) = \frac{(\rho_l - \rho_v)g\delta^2}{\mu_l}\left[\frac{y}{\delta} - \frac{1}{2}\left(\frac{y}{\delta}\right)^2\right]$$

[4 marks]

(c) Integrating again from 0 to δ to calculate the average:

$$\bar{u} = \frac{1}{\delta}\int_0^\delta u\,dy = \frac{1}{\delta}\int_0^\delta \frac{(\rho_l - \rho_v)g\delta^2}{\mu_l}\left[\frac{y}{\delta} - \frac{1}{2}\left(\frac{y}{\delta}\right)^2\right] dy$$

$$= \frac{(\rho_l - \rho_v)g\delta}{\mu_l}\int_0^\delta \left[\frac{y}{\delta} - \frac{1}{2}\left(\frac{y}{\delta}\right)^2\right] dy$$

$$= \frac{(\rho_l - \rho_v)g\delta}{\mu_l}\left[\frac{\delta^2}{2\delta} - \frac{\delta^3}{2\times 3\delta^2}\right] = \frac{(\rho_l - \rho_v)g\delta}{\mu_l}\left[\frac{\delta}{2} - \frac{\delta}{6}\right] = \frac{(\rho_l - \rho_v)g\delta}{\mu_l}\left[\frac{2\delta}{6}\right]$$

$$= \frac{(\rho_l - \rho_v)g\delta^2}{3\,\mu_l}$$

$$\delta(x) = \left[\frac{4k_l\mu_l(T_{sat} - T_w)x}{\rho_l(\rho_l - \rho_v)gh_{lv}}\right]^{\frac{1}{4}} = \left(\frac{4\,(680\times10^{-3})(279\times10^{-6})(100-95)0.3}{\frac{1000}{1.044}\left(\frac{1000}{1.044} - \frac{1}{1.679}\right)9.81(2257\times10^3)}\right)^{1/4} = 0.0000865 \text{ m}$$

$$\bar{u} = \frac{(\rho_l - \rho_v)g\delta^2}{3\,\mu_l} = \frac{\left(\frac{1000}{1.044} - \frac{1}{1.679}\right)9.81\times0.0000865^2}{3\times 279\times10^{-6}} = 0.084 \text{ m/s}$$

For very long plates, the thickness would increase and so the velocity. This could affect the conservation of laminar flow, becoming turbulent, where Nusselt theory won't apply

[8 marks]

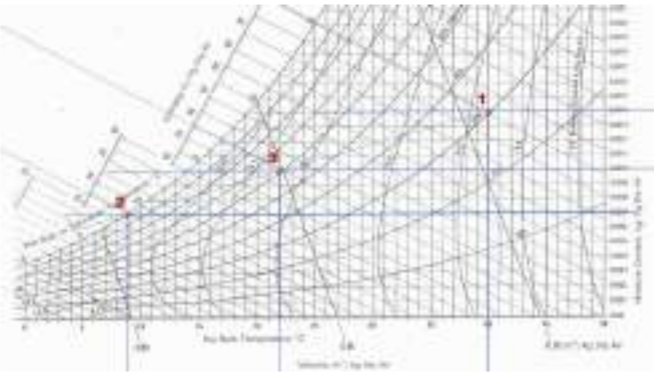
(d)

$$q = \frac{k_l\,(T_{sat} - T_w)}{\delta} = \frac{680\times10^{-3}(5)}{0.0000865} = 39306 \text{ W/m}^2$$

[4 marks]

QUESTION 4 **[17 points]**

(a)



(i) Using the psychrometric chart:

ω_1= 0.014 kg water (kg dry air)$^{-1}$

T_{wb}= 25ºC

T_{DP}= 19.2 ºC

[3 marks]

(ii) Using the chart, we can locate the specific humidity of stream 2, which should be saturated to use a minimum. We assume adiabatic

$m_{a1} = \frac{50}{(1+\omega_1)} = \frac{10}{(1+0.014)} = 49.31$ kg s^{-1}

ω_2: 0.007 kg water (kg dry air)$^{-1}$

ω_3: 0.010 kg water (kg dry air)$^{-1}$

$\dot{m}_{a1}\omega_1 + \dot{m}_{a2}\omega_2 = \dot{m}_{a3}\omega_3 = (\dot{m}_{a1} + \dot{m}_{a2})\omega_3$

$49.31 \times 0.014 + \dot{m}_{a2} \times 0.007 = (49.31 + \dot{m}_{a2})0.010$

$0.6903 + \dot{m}_{a2} \times 0.007 = 0.4931 + \dot{m}_{a2}0.010$

$\dot{m}_{a2} = \frac{0.4931 - 0.6903}{0.007 - 0.010} = 65.73$ kg s^{-1}

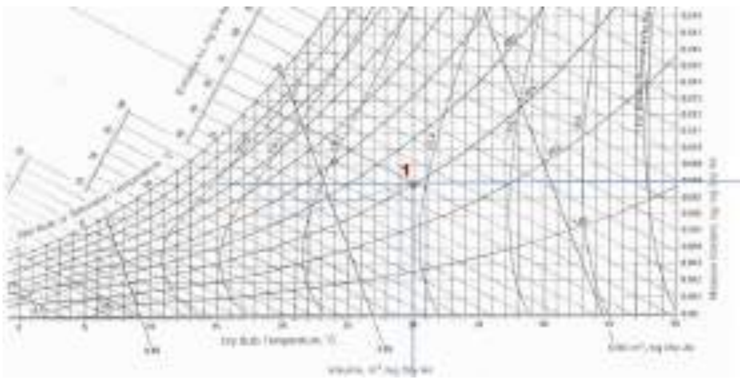
$\dot{m}_2 = 65.73\ (1 + 0.007) = 66.2$ kg s^{-1}

[5 marks]

(iii) Using the chart, we see that with evaporative cooling the stream would not be able to decrease its specific humidity, so it is not possible

[4 marks]

(b) Reading on graph, T_{wb}=18 ºC. It accounts for both sensible cooling and latent heat transfer (due to evaporation).



[5 marks]